

ACE INSTITUTE OF MANAGEMENT



**Pokhara University
Faculty of Management**



Bachelor in Computer System and Information Technology (BCSIT)

A PROJECT PROPOSAL ON

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Submitted by:

Group 4

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1. Introduction

In modern society, people frequently require help with small everyday tasks such as tutoring, home maintenance, basic technical support, delivery assistance, and other minor services. At the same time, many individuals possess valuable skills but lack a proper platform to showcase and monetize them within their local community.

Currently, people mostly rely on personal contacts, social media groups, or random online posts to find help. These methods are often unreliable, time-consuming, and lack transparency. Many skilled individuals also struggle to find opportunities to use their abilities and earn income from them. This results in unused skills within communities and difficulty in finding trusted service providers.

To address this gap, the Skill Exchange System also known as SkillSwap is proposed. SkillSwap is a web-based application designed to connect people who need help with individuals who can offer specific skills. The system functions as a matching platform similar to a dating application, where users create profiles, browse services, and connect with each other based on their needs.

Unlike traditional service platforms, SkillSwap focuses on local skill exchange and simple task assistance. Communication between users takes place through an in-app chat system, after which the service is performed offline. Service providers can decide their own pricing, while the platform earns revenue by charging a small commission on successful transactions.

The system aims to promote community collaboration, reduce the time required to find reliable help, and create a trustworthy environment for local service exchange.

2. Problem Statement

In today's fast-paced environment, individuals frequently encounter minor, everyday tasks ranging from household repairs to quick technical or creative fixes that they lack the time, tools, or specific expertise to handle themselves. Despite a large pool of skilled gig workers and independent contractors available locally, a severe friction point exists in the market.

The core issues holding this ecosystem back include:

i. Difficulty in Finding Reliable Help:

Service seekers face immense hurdles in locating dependable individuals who are willing to take on small-scale, short-term assignments.

ii. Lack of a Centralized Connection System:

There is currently no optimized, standard platform dedicated to seamlessly bridging the gap between local, skilled talent and the people who need them.

iii. Time Inefficiency:

Users waste an excessive amount of time relying on word-of-mouth, outdated directories, or fragmented social media groups to find help.

iv. Safety and Trust Risks:

Without a structured verification system, consumers are forced to rely on completely unknown or untrusted sources, creating security and quality assurance risks.

v. Inflated Service Costs:

Due to market fragmentation and a lack of pricing standards, users frequently end up paying disproportionately high costs for relatively simple services.

The Core Need

To resolve these inefficiencies, there is a critical need for an optimized, transparent system that connects local communities efficiently while ensuring absolute trust, safety, and operational transparency for both service seekers and providers.

3. Objectives of Study

The main objective of this project is to develop a platform that connects people who need services with skilled individuals who can provide them. Many workers such as plumbers, electricians, carpenters, tutors, and technicians have valuable skills but limited opportunities to reach customers. At the same time, customers often face difficulties finding reliable and skilled workers. This system aims to solve these problems by providing an easy-to-use platform where service seekers and service providers can communicate and interact efficiently.

i. To Connect Customers with Skilled Workers

The project aims to create a connection between customers and skilled workers. Many people spend a lot of time searching for reliable workers through friends, advertisements, or social media. The proposed system allows users to quickly find workers based on their skills, location, and availability.

Skilled individuals can create profiles to display their expertise and services, while customers can easily search and contact suitable workers. This improves job opportunities for workers and helps customers receive services more conveniently.

ii. To Promote Skill Sharing in the Community

Another objective is to encourage skill sharing within the community. Many talented individuals do not have a proper platform to showcase their abilities. This system provides them with opportunities to offer their services and gain recognition. The platform also supports local workers by helping them earn income and improve their professional growth. As a result, communities become more self-dependent by accessing local services easily.

iii. To Make Service Booking Easy and Fast

The project simplifies the traditional service booking process, which is often slow and inconvenient. Through the platform, users can browse services, compare workers, and book appointments quickly.

Features such as service categories, search filters, booking forms, and notifications make the process faster and more user-friendly. This saves time for both customers and workers while improving overall convenience.

iv. To Build Trust Through Ratings and Reviews

The system includes ratings and reviews to increase trust between customers and workers. After completing a service, customers can provide feedback based on their experience. Positive reviews help workers build a good reputation, while feedback encourages better service quality. This feature increases transparency and helps users make informed decisions before booking services.

v. To Manage Offline Services Efficiently

Although the platform is digital, most services are provided offline. Therefore, the system helps manage appointments, service requests, and communication between customers and workers effectively.

Once a booking is confirmed, both parties receive important details such as date, time, and location. The platform also stores service records for future references, making the process more organized and reliable.

4. Methodology

4.1. Requirement Identification

Requirement identification is the process of understanding the problems, user needs, and system requirements before developing the system. In this project, requirement identification was performed by studying existing systems and identifying the needs of users.

4.1.1. Study of Existing Systems

Existing platforms such as Skillshare, Skillpact and Skilljar were studied to understand how skill-sharing systems work and what features they provide.

These systems mainly focus on online learning and paid courses. They provide features such as:

- User registration and login
- Skill/course listing
- Communication features
- Rating and review systems

However, these platforms have some limitations:

- Limited local community interaction
- Focus mainly on online learning
- Lack of flexible offline meetup support
- Limited peer-to-peer real-world service exchange

Based on this study, the proposed system aims to provide a more community-oriented platform where users can connect locally, communicate directly, schedule services, and exchange skills through offline interaction.

Several existing studies and platforms related to peer-to-peer learning and skill exchange were analyzed during the requirement identification phase.

A study by Sri (2025) highlighted the importance of user profiles, communication systems, and collaborative learning in skill-sharing applications. Similarly, Jadhav et

al. (2025) focused on matchmaking and community interaction in peer-to-peer skill exchange platforms.

Another study conducted by Anam et al. (2025) presented a web-based collaborative learning platform developed using PHP and MySQL, providing features such as authentication and skill management. Furthermore, Ukata and Moko (2026) emphasized peer-to-peer interaction and practical learning through mobile-based skill exchange applications.

These studies helped identify important system requirements such as communication, booking management, user profiles, and community-based interaction for the proposed system.

4.1.2. Identify User Needs

The system is designed to fulfill the needs of both service providers and service seekers.

The identified user needs are:

- Users should be able to create accounts and manage profiles
- Service providers should be able to list skills and service pricing
- Service seekers should be able to search for skills and nearby providers
- Users should communicate through an in-app chat system
- Users should be able to send and manage booking requests
- The system should support payment processing
- The platform should include ratings and reviews to build trust
- Users should have a dashboard to manage activities and bookings

These requirements help ensure that the system is practical, user-friendly, and efficient.

4.2. Feasibility Study

A feasibility study was conducted to determine whether the proposed system can be developed successfully within available resources, technology, and time.

4.2.1. Technical Feasibility

Our system is technically feasible because it can be developed using commonly available web technologies.

The technologies planned for development include:

- Frontend: HTML, CSS, JavaScript, React.js
- Backend: PHP
- Database: MySQL

The required tools, frameworks, and development environments are accessible and suitable for a student-level project. The development team has basic knowledge of these technologies, making implementation achievable.

4.2.2. Operational Feasibility

The system is operationally feasible because it is designed to be simple and user-friendly.

Users can easily:

- Register and create profiles
- Search for services
- Communicate with providers
- Book services
- Make payments and provide feedback

The platform supports practical real-world interaction and addresses the problem of finding trusted local services efficiently.

4.2.3. Economic Feasibility

The project is economically feasible because the development cost is relatively low.

Most of the required technologies and tools are open-source and freely available. The project mainly requires:

- Internet access
- Development tools
- Basic hosting services

Since the system is developed as an academic project, no major financial investment is required. Additionally, the platform can generate revenue in the future through commission-based payments from service transactions.

4.3. High Level Design of System

The high-level design describes the overall structure, workflow, and major components of the proposed Skill Exchange System. It explains how users interact with the system and how different modules work together to perform the required functionalities.

4.3.1. System Flow

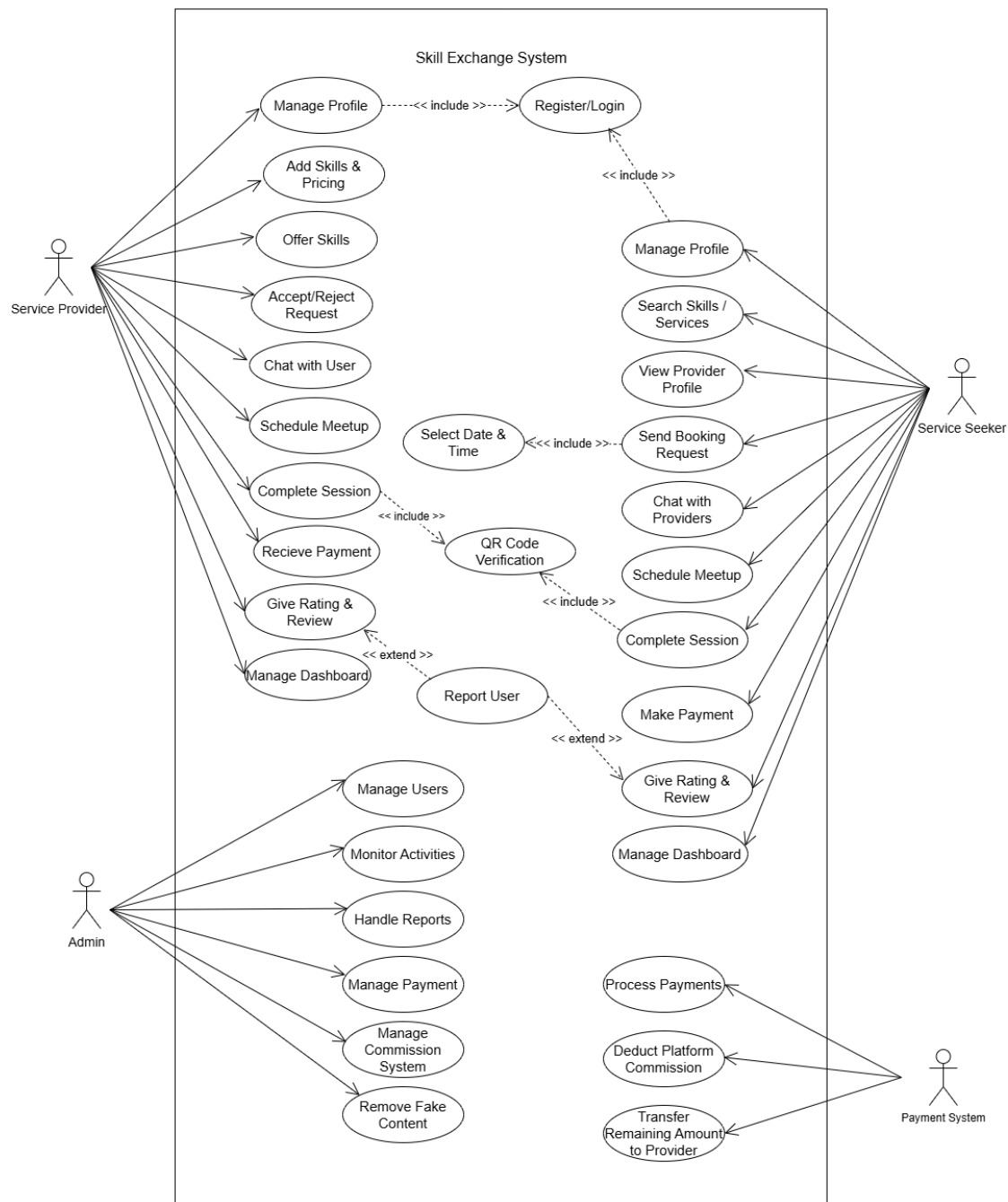


Figure 1: Use Case Diagram of Skill Exchange System

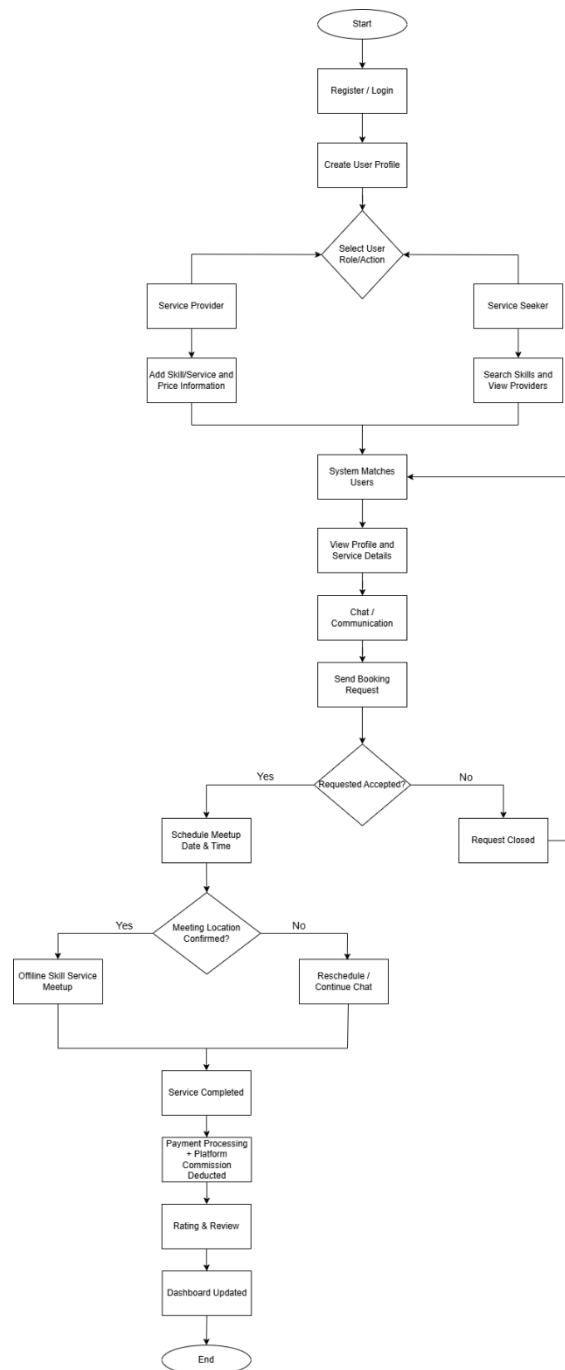


Figure 2: Flowchart of Skill Exchange System

The system flowchart represents the overall workflow of the platform from user registration to service completion.

The process begins when users register and create profiles. Service providers add skills and pricing information, while service seekers search for services and providers. The system matches users based on their requirements and enables communication through the chat system.

After communication, booking requests are sent and managed by the provider. Once accepted, users schedule an offline meetup to complete the service. After successful completion, payment is processed, platform commission is deducted, and ratings and reviews are provided.

The flowchart helps visualize the step-by-step operation of the entire system.

4.3.2. Basic Architecture

This system follows a three-tier web application architecture consisting of:

I. Presentation Layer (Frontend)

This layer handles user interaction and interface design. Users interact with the system through web pages developed using:

- HTML
- CSS
- JavaScript
- React.js

This layer includes:

- Login pages
- Dashboard
- Skill listing pages
- Booking interfaces
- Chat interfaces

II. Application Layer (Backend)

The backend processes business logic and system operations. It manages:

- User authentication
- Booking management
- Chat handling
- Payment processing
- Rating and review management

The backend will be developed using:

- PHP

III. Database Layer

This layer stores and manages all system data including:

- User information
- Skills and services
- Bookings
- Payments
- Ratings and reviews

The database system used is:

- MySQL

4.3.3. Working Mechanism

The working mechanism explains how the system functions during actual usage.

- i. Users register and log into the system.
- ii. Service providers add their skills, service details, and pricing.
- iii. Service seekers search for required skills or nearby providers.
- iv. Users communicate using the chat system.
- v. Service seekers send booking requests.
- vi. Service providers accept or reject requests.
- vii. After acceptance, both users schedule an offline meetup.
- viii. The service is completed physically.
- ix. Payment is processed through the system.
- x. Platform commission is deducted automatically.
- xi. Users provide ratings and reviews after service completion.

This mechanism ensures smooth communication, organized booking management, secure transactions, and trust between users.

5. Gantt Chart

5.1. Project Timeline

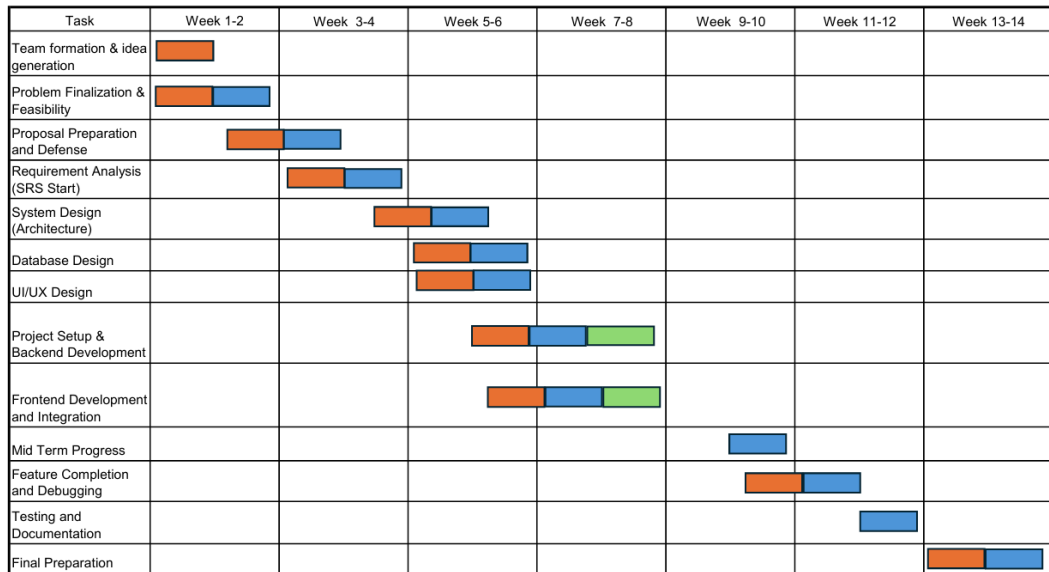


Figure 3: Gantt Chart of the Project

5.2. Task Distribution

The project will be developed collaboratively by four team members. Each member will be assigned specific responsibilities based on different development areas to ensure efficient project management and balanced workload distribution.

Anubhav Dahal – UI/UX Design, Frontend and Backend Coordination

- Design the overall UI/UX of the system
- Create wireframes and interface layouts
- Assist in frontend development
- Coordinate frontend and backend integration
- Support backend implementation when required

Grishma Adhikari – Frontend Development

- Develop user interface components using frontend technologies
- Implement responsive web pages

- Assist in dashboard and user interaction development
- Work on integrating frontend features with backend APIs

Krish Shrestha– Frontend Development

- Support frontend design and implementation
- Develop pages related to booking, profiles, and communication
- Assist in testing frontend functionality
- Improve user experience and interface consistency

Sulav Raj Shrestha– Backend Development

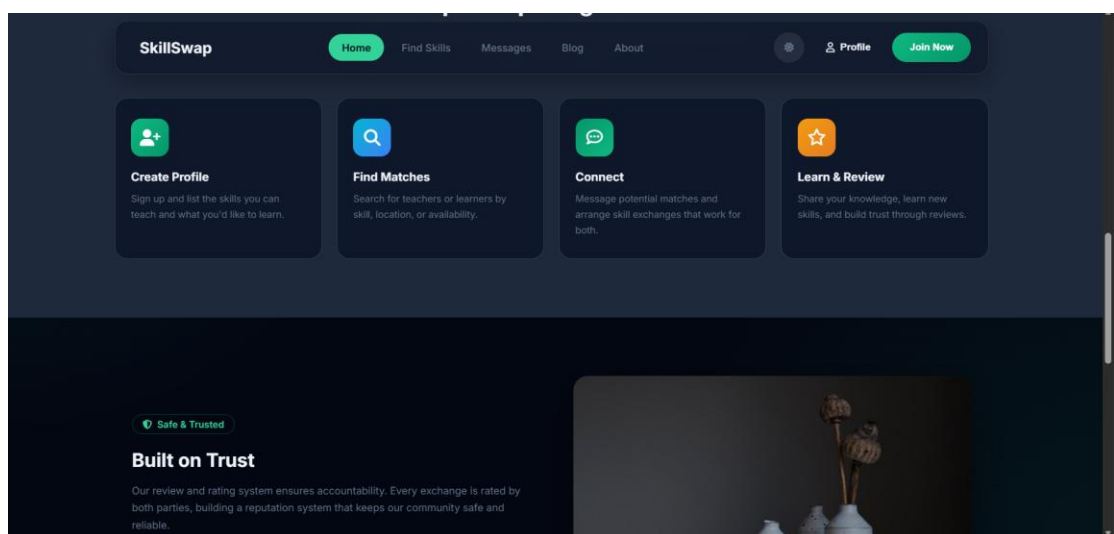
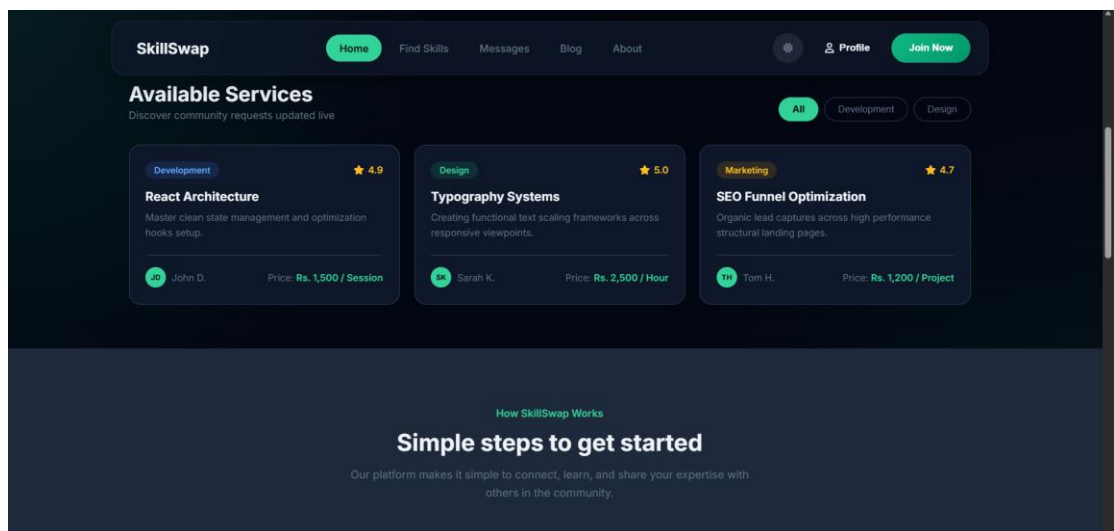
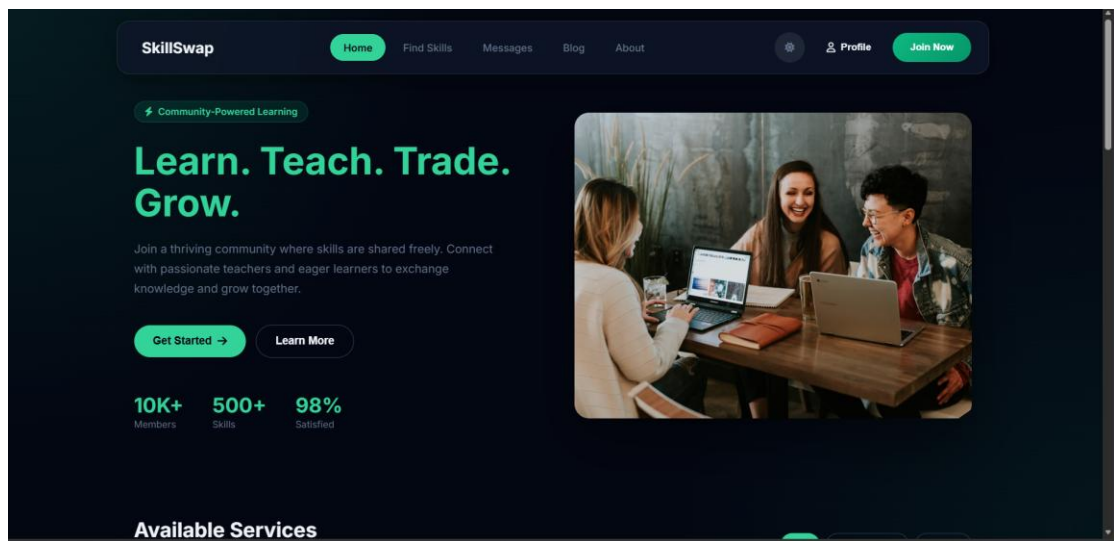
- Develop backend logic and server-side functionality
- Manage database integration and system operations
- Implement authentication, booking, payment, and commission handling
- Ensure system security and backend performance

6. Expected Outcome

The expected outcome of this project is the successful development of a web-based Skill Exchange System that connects service seekers with skilled individuals in a simple, reliable, and organized way. The platform will provide an easy method for users to search for local services, communicate with service providers, and manage bookings efficiently.

The final result of the project will be a functional prototype of the Skill Exchange System that promotes community collaboration, reduces the difficulty of finding trusted local services, and creates better opportunities for skilled individuals to earn income through their abilities.

7. Interface Design



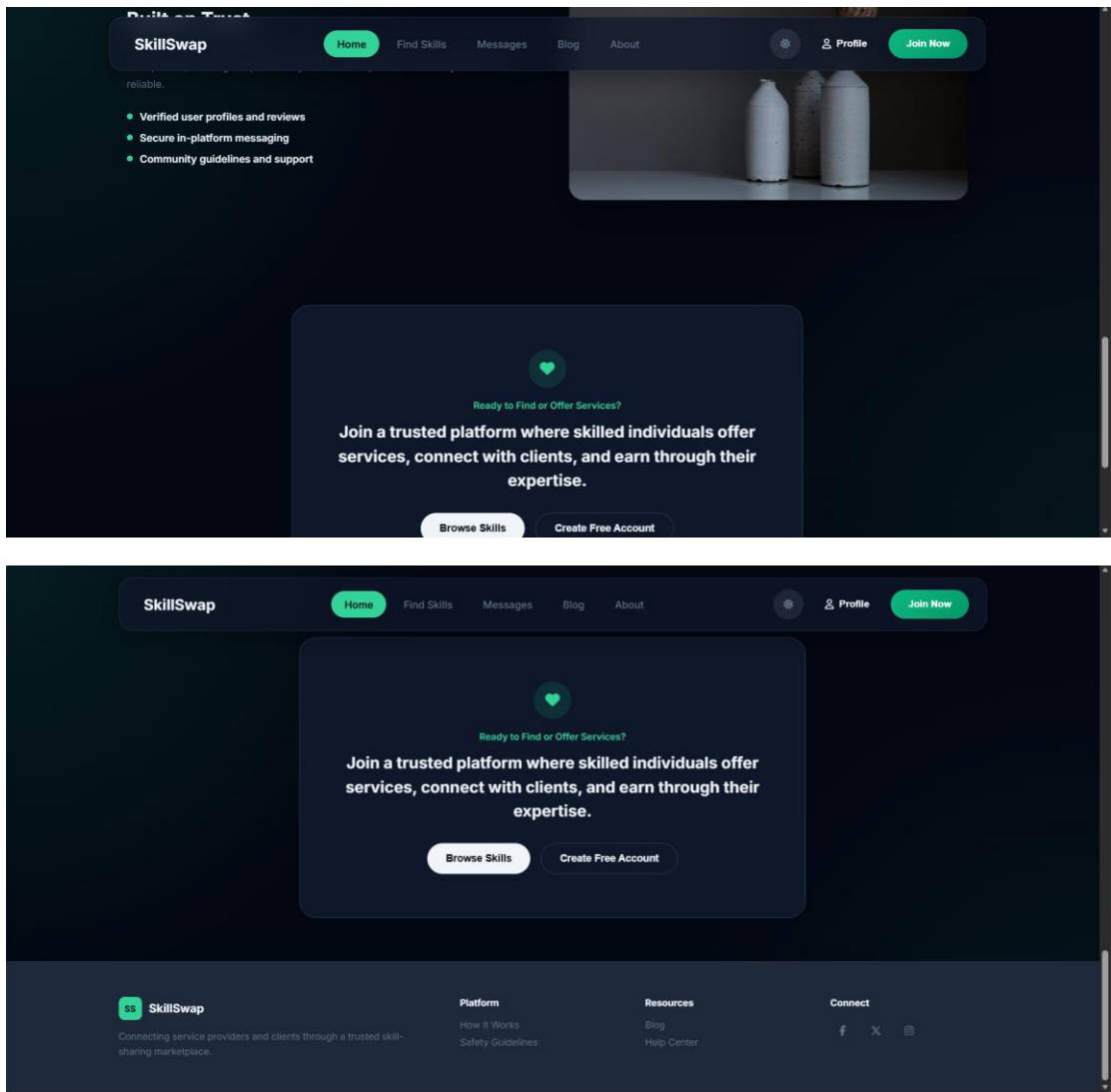
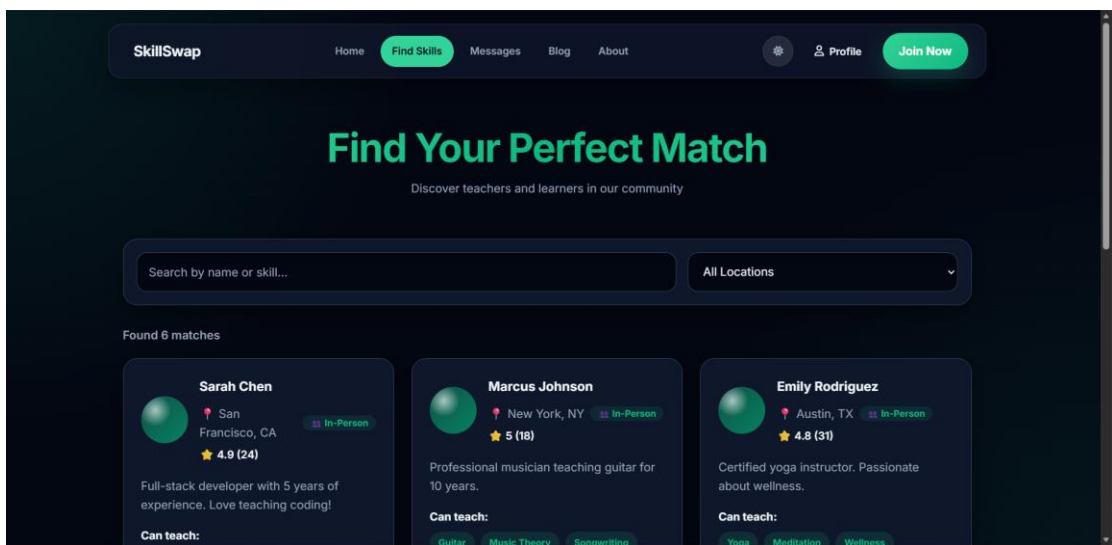


Figure 1: Home Page



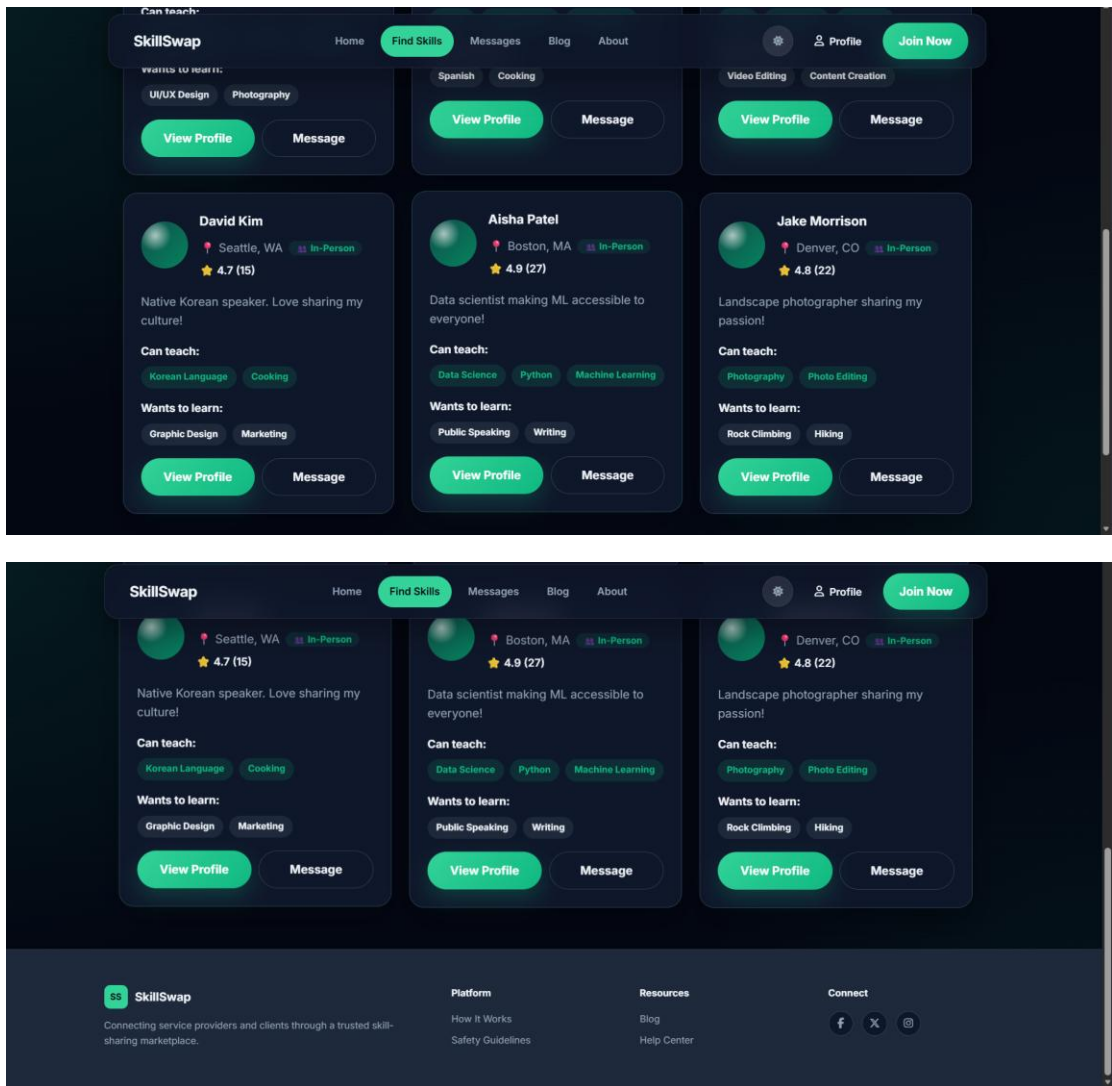
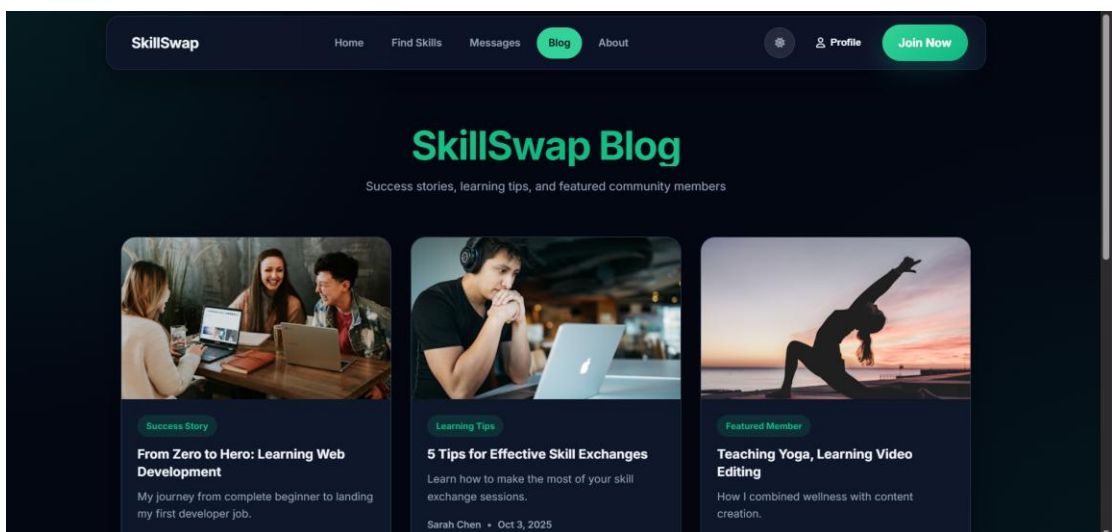


Figure 2: Find Skills Page



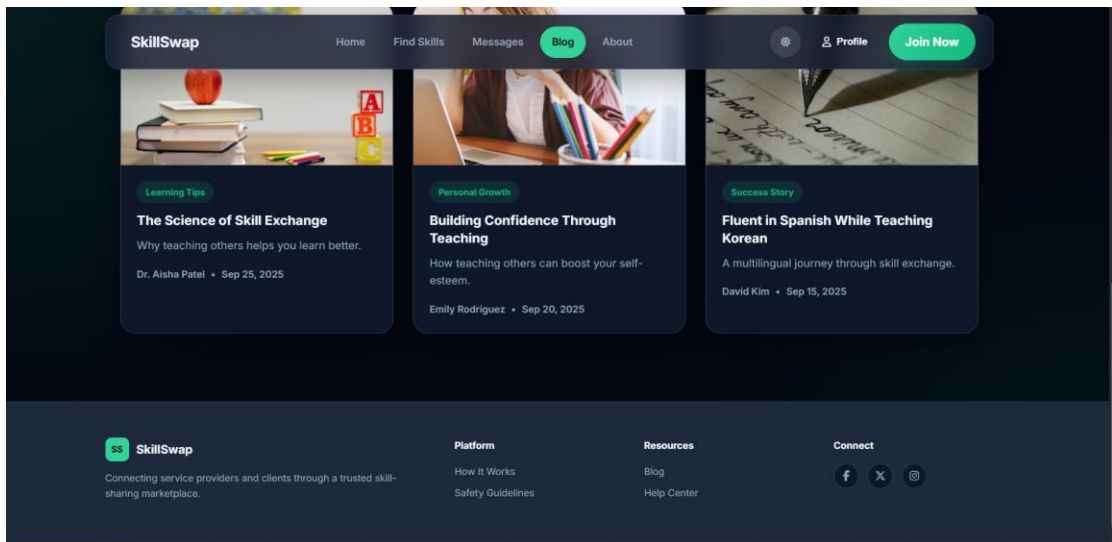
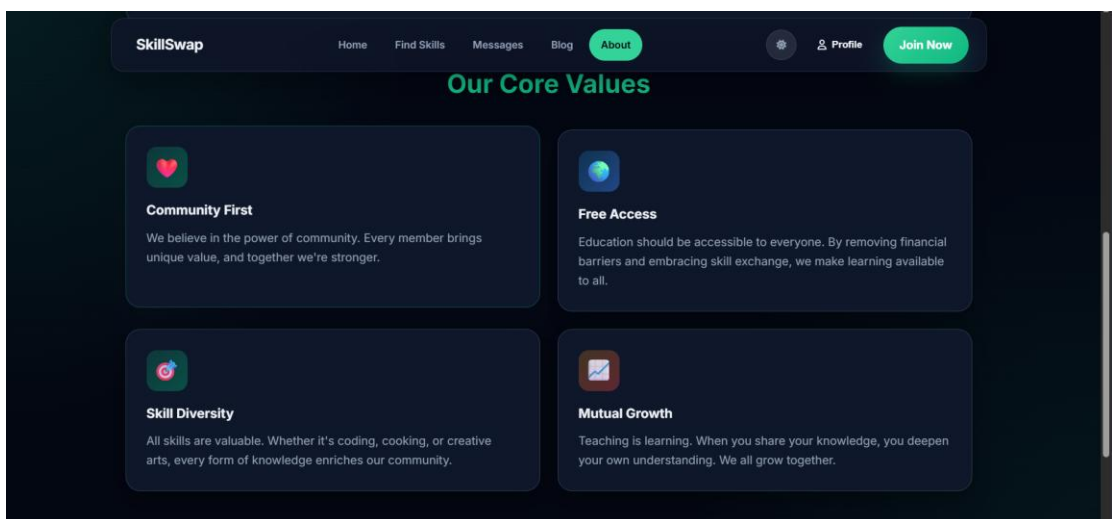
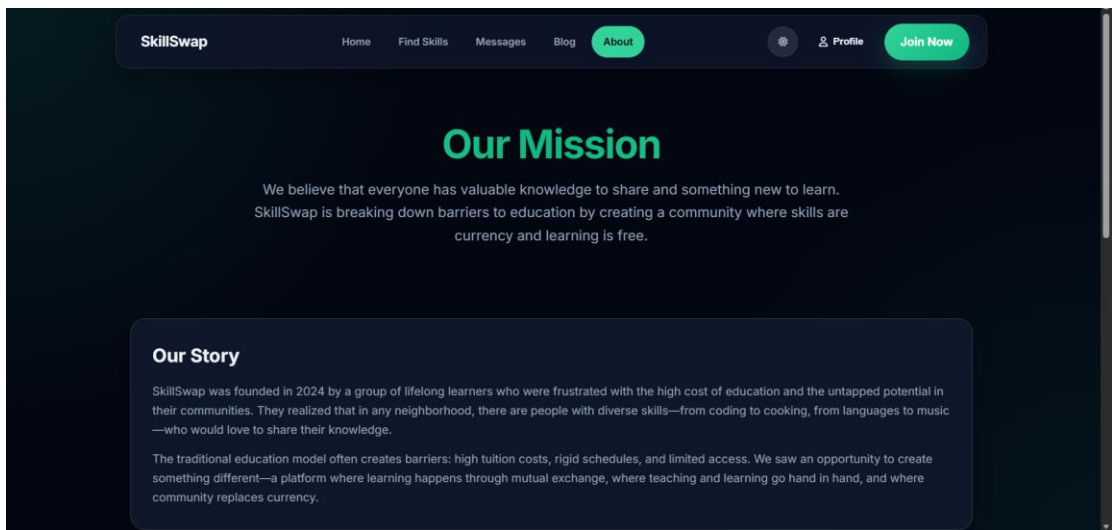


Figure 3: Blog Page



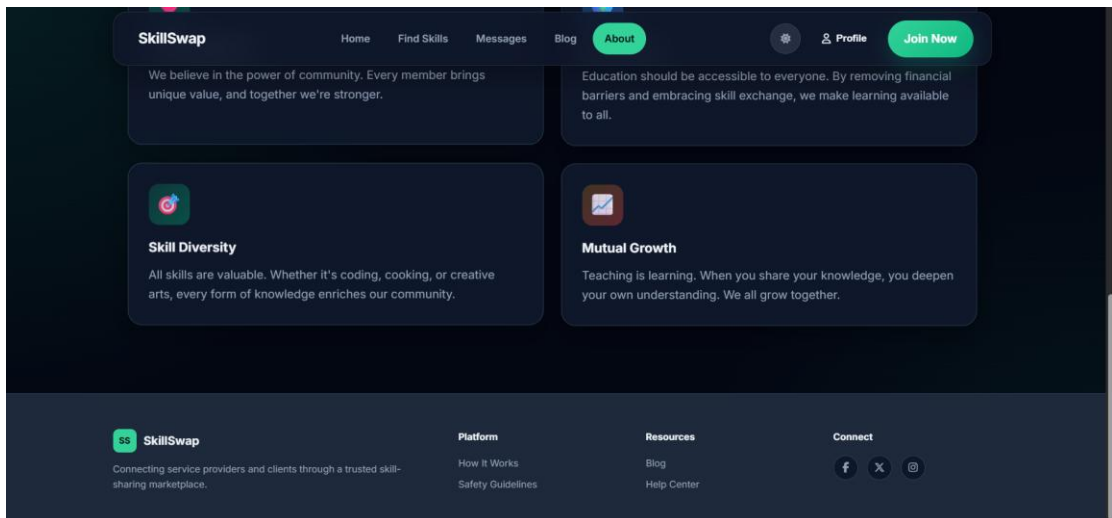


Figure 4: About Page

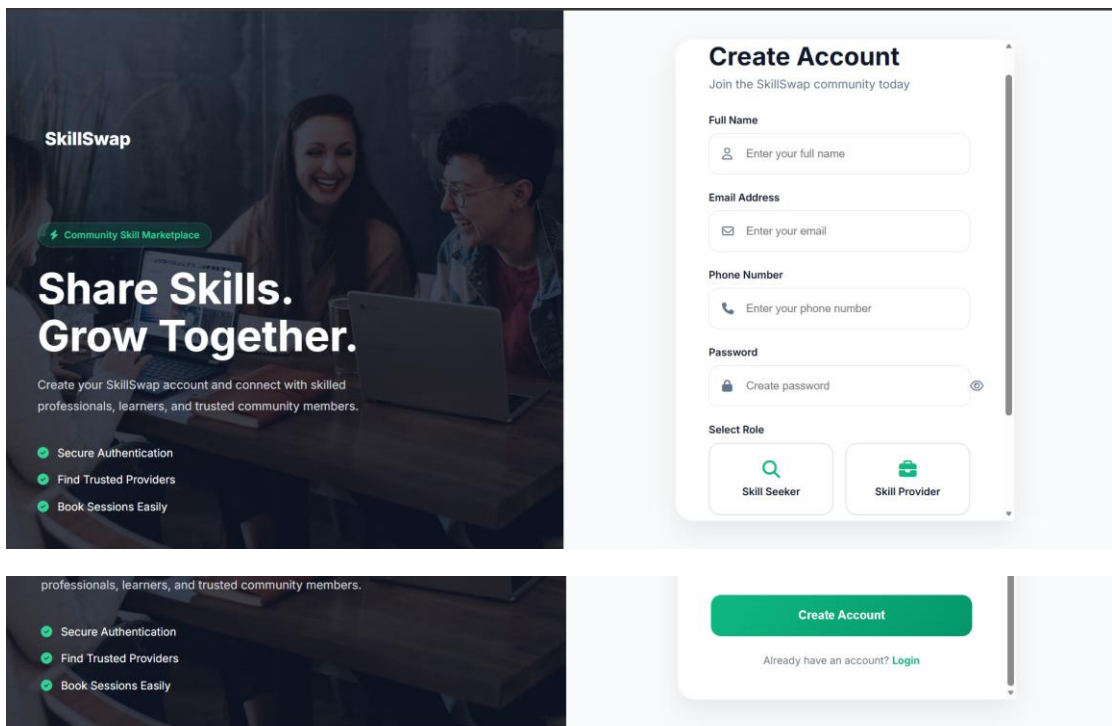


Figure 5: Sign up Page

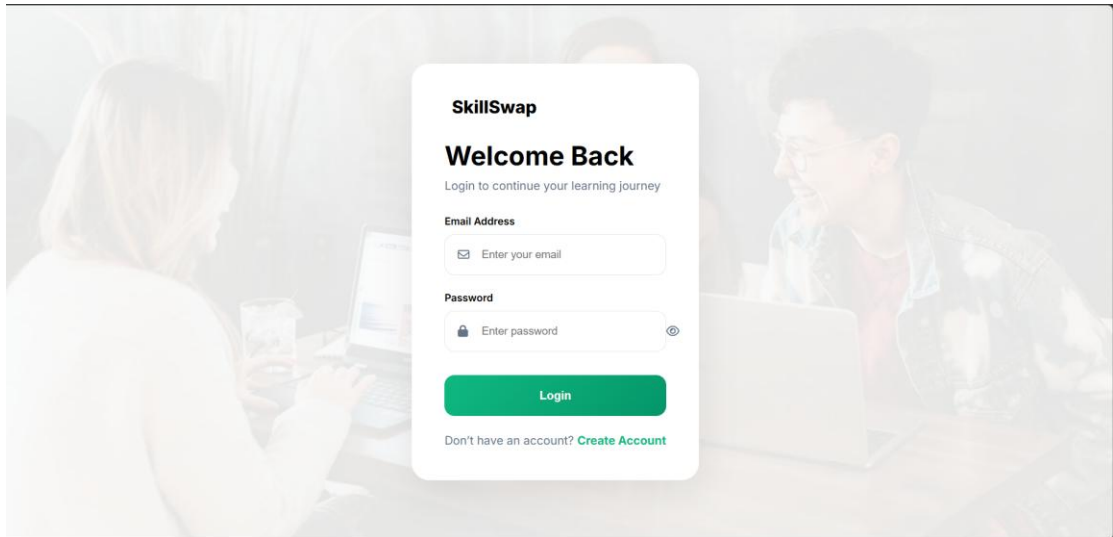


Figure 6: Login Page

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